ECE 332: Engineering Electromagnetics II

Lecture 8

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April 23, 2026

Quick Review

- The **phasor of a plane wave** propagating **in a lossy medium** satisfies the wave equation

$$\nabla^2 \tilde{\mathbf{E}} - \gamma^2 \tilde{\mathbf{E}} = \mathbf{0}, \tag{1}$$

where

$$\gamma^2 = -\omega^2 \mu \epsilon_c \quad \text{and} \quad \epsilon_c = \epsilon' - j\epsilon'' = \epsilon - j\frac{\sigma}{\omega}. \tag{2}$$

- The **attenuation constant** α is the real part of γ and the **phase constant** β is the imaginary part of γ. These are given by

$$\alpha = \omega \left\{ \frac{\mu\epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'}\right)^2} - 1 \right] \right\}^{1/2} \quad \text{(Np/m)} \tag{3}$$

$$\beta = \omega \left\{ \frac{\mu\epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'}\right)^2} + 1 \right] \right\}^{1/2} \quad \text{(rad/m)} \tag{4}$$

(Note: Np = nepers, which is a logarithmic scale for ratios, like dB.)

Quick Review (Continued)

- Assuming there is only a wave traveling in the $+z$ direction, the **electric field phasor satisfying the wave equation** is

$$\tilde{\mathbf{E}}(z) = E_{x0} e^{-\alpha z} e^{-j\beta z} \hat{\mathbf{x}} \tag{5}$$

and the corresponding **magnetic field phasor** is

$$\tilde{\mathbf{H}}(z) = \frac{E_{x0}}{\eta_c} e^{-\alpha z} e^{-j\beta z} \hat{\mathbf{y}}. \tag{6}$$

- The **intrinsic impedance for the lossy medium** is

$$\eta_c = \sqrt{\frac{\mu}{\epsilon_c}} = \sqrt{\frac{\mu}{\epsilon'}} \left(1 - j\frac{\epsilon''}{\epsilon'}\right)^{-1/2}, \tag{7}$$

where the **complex permittivity** is

$$\epsilon_c = \epsilon' - j\epsilon'' = \epsilon - j\frac{\sigma}{\omega}. \tag{8}$$

Since η_c is complex, $\tilde{\mathbf{E}}(z)$ and $\tilde{\mathbf{H}}(z)$ (and $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$) **do not have the same phase**.

Quick Review (Continued)

- The **magnitudes**

$$|\tilde{\mathbf{E}}(z)| = |E_{x0}|e^{-\alpha z}, \quad |\tilde{\mathbf{H}}(z)| = \left|\frac{E_{x0}}{\eta_c}\right| e^{-\alpha z} \tag{9}$$

  decrease exponentially at a rate α and decrease by a factor of $1/e$ for each multiple of the **skin depth**

$$\delta_s = 1/\alpha \text{ m}. \tag{10}$$

  The lost energy is dissipated as heat.

- The **expressions for** α, β, and η_c are correct for any **linear, isotropic, homogeneous medium**.

- Media are characterized by the ratio ϵ''/ϵ':
  - $\epsilon''/\epsilon' \ll 1$ – **low-loss dielectric**
  - $\epsilon''/\epsilon' \gg 1$ – **good conductor**
  - $10^{-2} \leq \epsilon''/\epsilon' \leq 10^{2}$ – **quasi-conductor**

Quick Review (Continued)

- **Low-Loss Dielectrics**

  Here, $\epsilon''/\epsilon' \ll 1$, which implies

$$\alpha \approx \frac{\omega\epsilon''}{2}\sqrt{\frac{\mu}{\epsilon'}} = \frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon}} \ \ (\text{Np/m}), \tag{11}$$

$$\beta \approx \omega\sqrt{\mu\epsilon'} = \omega\sqrt{\mu\epsilon} \ \ (\text{rad/m}), \tag{12}$$

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \ \ (\Omega). \tag{13}$$

  So, in this case,

  - the **phase constant** β is the same as the **wavenumber** k **for a lossless medium**,
  - the **intrinsic impedance** η_c is the same as the **intrinsic impedance** η **of a lossless medium**.

Quick Review (Continued)

- **Good Conductors**

  In this case, $\epsilon''/\epsilon' > 100$ and

$$\alpha \approx \omega\sqrt{\frac{\mu\epsilon''}{2}} = \omega\sqrt{\frac{\mu\sigma}{2\omega}} = \sqrt{\pi f \mu\sigma} \quad \text{(Np/m)}, \tag{14}$$

$$\beta \approx \alpha \approx \sqrt{\pi f \mu\sigma} \quad \text{(rad/m)}, \tag{15}$$

$$\eta_c \approx \sqrt{j\frac{\mu}{\epsilon''}} = (1+j)\sqrt{\frac{\pi f \mu}{\sigma}} \approx (1+j)\frac{\alpha}{\sigma} \quad (\Omega). \tag{16}$$

- We explored an example of finding $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$ and the depth z where $|\mathbf{E}(z)| = 0.01|\mathbf{E}(0)|$ for a uniform plane wave in seawater.

Quick Review (Continued)

Current Flow in a Good Conductor

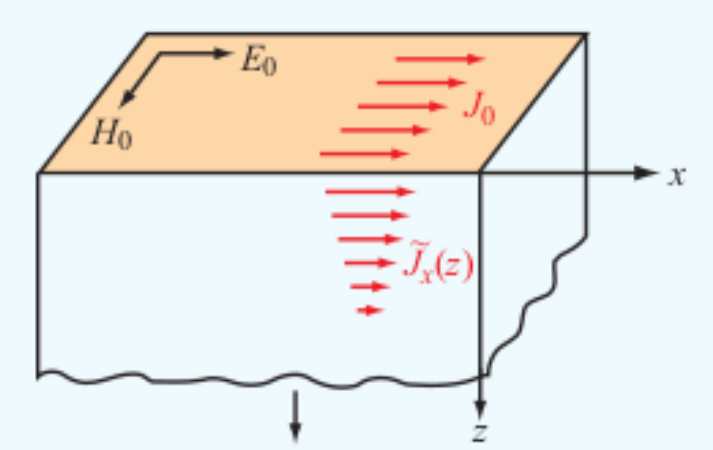
- The **current** associated with a **DC voltage source** is **evenly distributed** throughout a conductor.
- In contrast, the **current** associated with an **AC voltage source** mostly **flows through a layer near the surface**.

Consider a semi-infinite conductor bounded by a perfect dielectric at the x-y plane with the z axis pointing into the conductor.

- An x-**polarized electric field**

$$\tilde{\mathbf{E}} = E_0 \hat{\mathbf{x}} \tag{17}$$

at the surface



(a) Exponentially decaying $\tilde{J}_x(z)$

Quick Review (Continued)

induces a plane wave

$$\tilde{\mathbf{E}}(z) = E_0 e^{-\alpha z} e^{-j\beta z} \hat{\mathbf{x}} \tag{18}$$

$$\tilde{\mathbf{H}}(z) = \frac{E_0}{\eta_c} e^{-\alpha z} e^{-j\beta z} \hat{\mathbf{y}} \tag{19}$$

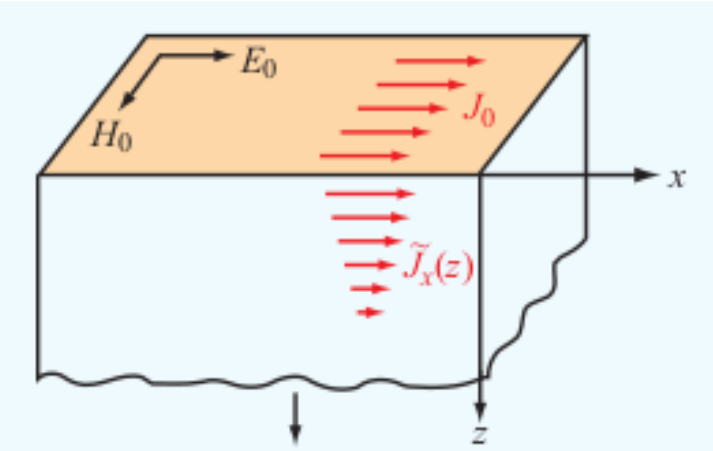
in the conductor. This creates a current

$$\begin{aligned} \tilde{\mathbf{J}}(z) &= J_0 e^{-\alpha z} e^{-j\beta z} \hat{\mathbf{x}} \\ &= J_0 e^{-(1+j)z/\delta_s} \hat{\mathbf{x}} \ (\mathrm{A/m^2}), \end{aligned} \tag{20}$$

where

$$J_0 = \sigma E_0 \tag{21}$$

is the **current density in the conductor at the boundary**.



(a) Exponentially decaying $\tilde{J}_x(z)$

Quick Review (Continued)

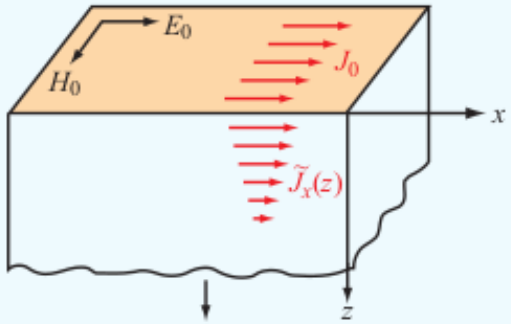
The **skin depth**

$$\delta_s = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f \mu \sigma}} \text{ (m)}. \qquad (22)$$

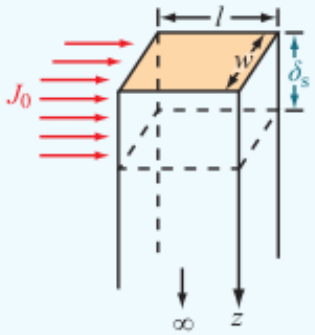
- The **total current passing through a slice of the conductor with width w in the y direction** is

$$\tilde{I} = \frac{J_0 w \delta_s}{1+j}. \qquad (23)$$

This is an excellent approximation if the depth is finite $>$ a few times δ_s.



(a) Exponentially decaying $\tilde{J}_x(z)$



(b) Equivalent J_0 over skin depth δ_s

Figure 7-15 Exponential decay of current density $\tilde{J}_x(z)$ with z in a solid conductor. The total current flowing through (a) a section of width w extending between $z = 0$ and $z = \infty$ is equivalent to (b) a constant current density J_0 flowing through a section of depth δ_s.

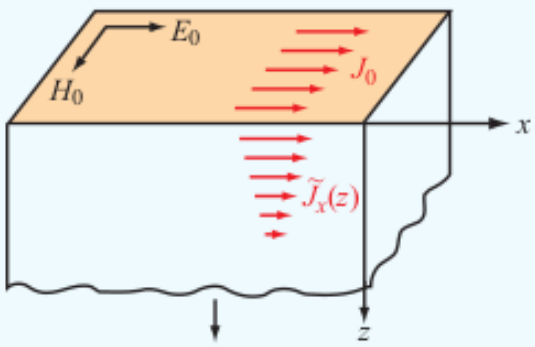
Current Flow in a Good Conductor (Continued)

If the slice has thickness ℓ (in the x direction), the **voltage across the slice at the surface** is

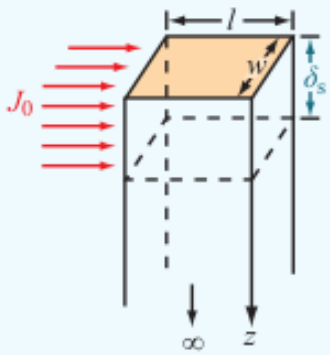
$$\tilde{V} = E_0 \ell = \frac{J_0}{\sigma} \ell \tag{24}$$

and the impedance of the slice is

$$\begin{aligned} Z = \frac{\tilde{V}}{\tilde{I}} &= \left(\frac{J_0 \ell}{\sigma}\right)\left(\frac{1+j}{J_0 w \delta_s}\right) \\ &= \left(\frac{1+j}{\sigma \delta_s}\right)\frac{\ell}{w} \\ &= Z_s \frac{\ell}{w} \ (\Omega), \end{aligned} \tag{25}$$



(a) Exponentially decaying $\tilde{J}_x(z)$



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Current Flow in a Good Conductor (Continued)

where

$$Z_s = \frac{1+j}{\sigma\delta_s} \tag{26}$$

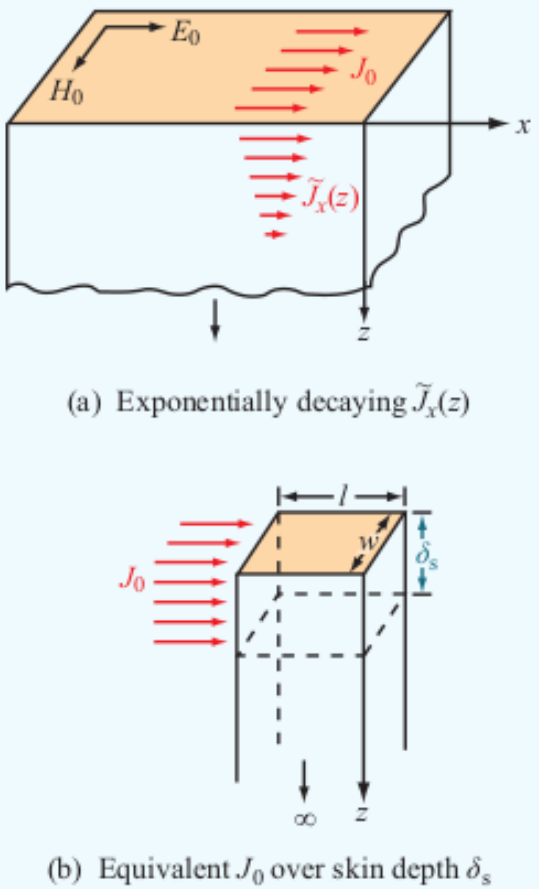
is the **internal** or **surface impedance**. (Z_s is the impedance for a slice with a square cross-section (i.e. with $\ell = w$).)

Since the real part of Z_s is positive,

$$Z_s = R_s + j\omega L_s \tag{27}$$

with **surface resistance**

$$R_s = \frac{1}{\sigma\delta_s} = \frac{\alpha}{\sigma} = \frac{\sqrt{\pi f \mu \sigma}}{\sigma}$$
$$= \sqrt{\frac{\pi f \mu}{\sigma}}\ (\Omega) \tag{28}$$



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Current Flow in a Good Conductor (Continued)

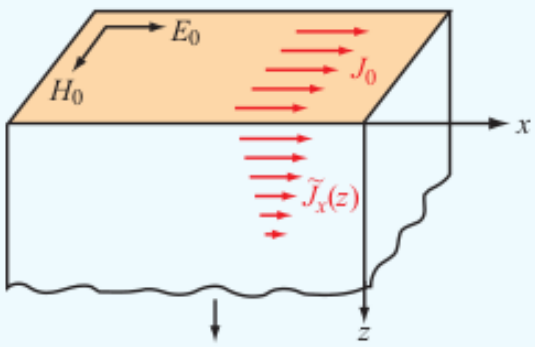
and inductance

$$L_s = \frac{1}{\omega\sigma\delta_s} = \frac{\sqrt{\pi f \mu \sigma}}{2\pi f \sigma}$$
$$= \frac{1}{2}\sqrt{\frac{\mu}{\pi f \sigma}} \text{ (H)}. \qquad (29)$$

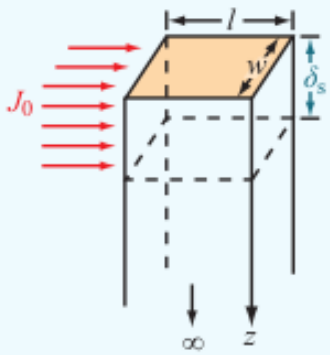
The **ac resistance of a slab of width** w **and length** ℓ is

$$R = R_s \frac{\ell}{w} = \frac{\ell}{\sigma \delta_s w}, \qquad (30)$$

which is the same as the dc resistance of a plane conductor of length ℓ and cross-section $A = \delta_s w$.



(a) Exponentially decaying $\tilde{J}_x(z)$



(b) Equivalent J_0 over skin depth δ_s

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Current Flow in a Good Conductor (Continued)

The above results can be applied to a coaxial cable that has a cylindrical inner conductor with a radius > several times the skin depth δ_s.

It's easy to produce a cable that meets this criterion. For example, the skin depth for a wave with frequency $f = 1$ MHz in a copper conductor is

$$\delta_s = \frac{1}{\sqrt{\pi f \mu \sigma}} = 0.066 \text{ mm} \qquad (31)$$

and is smaller for higher frequencies.

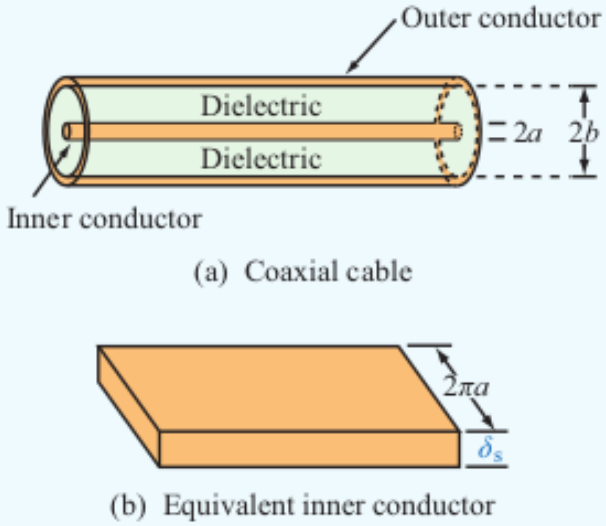


Figure 7-16 The inner conductor of the coaxial cable in (a) is represented in (b) by a planar conductor of width $2\pi a$ and depth δ_s, as if its skin has been cut along its length on the bottom side and then unfurled into a planar geometry.

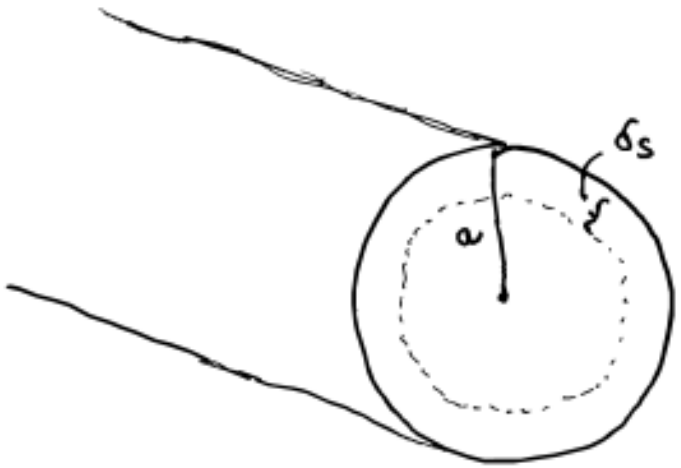
Current Flow in a Good Conductor (Continued)

Since most of the current flows through the portion of the inner conductor that extends from the surface to a depth δ_s, the inner conductor is roughly equivalent to a slab of width w equal to the circumference of the cylinder $2\pi a$ and depth δ_s. So, the resistance per unit length is

$$R_1' = \frac{R}{\ell} = \frac{R_s}{2\pi a} \ (\Omega/\text{m}). \qquad (32)$$

Similarly, the resistance per unit length for the outer conductor with radius b is

$$R_2' = \frac{R_s}{2\pi b} \ (\Omega/\text{m}). \qquad (33)$$



(b) Equivalent inner conductor

Current Flow in a Good Conductor (Continued)

Altogether, the **ac resistance of a coaxial cable per unit length** is

$$R' = R_1' + R_2' = \frac{R_s}{2\pi}\left(\frac{1}{a} + \frac{1}{b}\right) \ (\Omega/\text{m}). \tag{34}$$

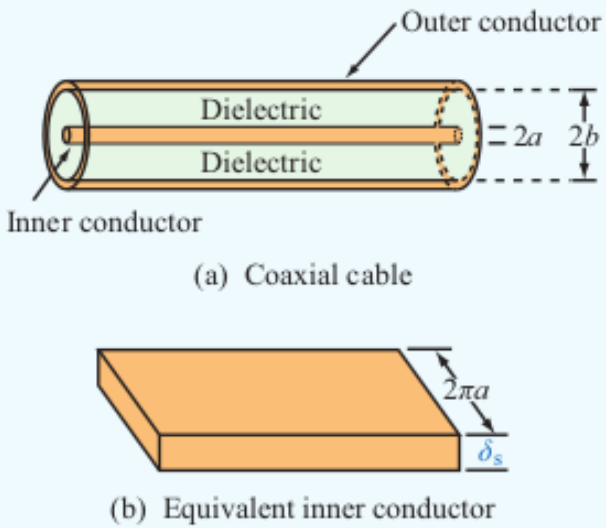


Figure 7-16 The inner conductor of the coaxial cable in (a) is represented in (b) by a planar conductor of width $2\pi a$ and depth δ_s, as if its skin has been cut along its length on the bottom side and then unfurled into a planar geometry.

Electromagnetic Power Density

How much power does an electromagnetic wave carry?

The **Poynting vector** is

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad (\mathrm{W/m^2}), \tag{35}$$

which has units of power per unit area. Notice that **S** is perpendicular to **E** and **H** and, in fact, points in the direction of propagation of the EM wave.

If the wave passes through an area A with unit surface vector $\hat{\mathbf{n}}$, then the power flowing through the area A is

$$P = \int_A \mathbf{S} \cdot \hat{\mathbf{n}}\, dA \quad (\mathrm{W}) \tag{36}$$

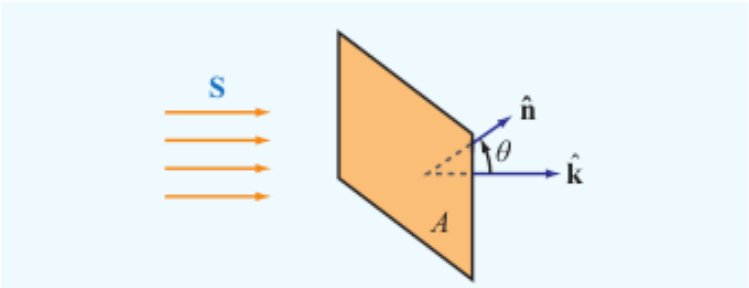


Figure 7-17 EM power flow through an aperture.

If the wave propagates in direction $\hat{\mathbf{k}}$, $S = |\mathbf{S}|$, and θ is the angle between $\hat{\mathbf{k}}$ and $\hat{\mathbf{n}}$

$$P = SA\cos\theta \quad (\mathrm{W}). \tag{37}$$

Electromagnetic Power Density (Continued)

The Poynting vector **S** is a function of time since **E** and **H** are functions of time. So, the **average power density**

$$\mathbf{S}_{\text{av}} = \frac{1}{2}\,\text{Re}\left\{\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*\right\} \quad (\text{W/m}^2) \tag{38}$$

is a better measure of the power the wave carries.

Note that **S** is analogous to the **instantaneous power**

$$P(z,t) = v(z,t)i(z,t) \tag{39}$$

flowing through a transmission line, where $v(z,t)$ and $i(z,t)$ are the **instantaneous voltage and current**, respectively, and $\mathbf{S}_{\text{av}}$ is analogous to the **time-average power**

$$P_{\text{av}}(z) = \frac{1}{2}\,\text{Re}\left\{\tilde{V}(z)\tilde{I}^*(z)\right\} \tag{40}$$

that a transmission line carries, where $\tilde{V}(z)$ and $\tilde{I}(z)$ are the phasors of $v(z,t)$ and $i(z,t)$.

Electromagnetic Power Density (Continued)

Power of an EM Wave in a Lossless Medium

For a uniform plane wave traveling in the $+z$ direction,

$$\tilde{\mathbf{E}}(z) = (E_{x0}\hat{\mathbf{x}} + E_{y0}\hat{\mathbf{y}})e^{-jkz} \tag{41}$$

and

$$\begin{aligned}
\tilde{\mathbf{H}}(z) &= \frac{1}{\eta}\hat{\mathbf{z}} \times \tilde{\mathbf{E}} = \frac{1}{\eta}\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ 0 & 0 & 1 \\ E_{x0}e^{-jkz} & E_{y0}e^{-jkz} & 0 \end{vmatrix} \\
&= \frac{1}{\eta}\left(-E_{y0}e^{-jkz}\hat{\mathbf{x}} + E_{x0}e^{-jkz}\hat{\mathbf{y}}\right) \\
&= \frac{1}{\eta}\left(-E_{y0}\hat{\mathbf{x}} + E_{x0}\hat{\mathbf{y}}\right)e^{-jkz}.
\end{aligned} \tag{42}$$

Electromagnetic Power Density (Continued)

So,

$$\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^* = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ E_{x0} e^{-jkz} & E_{y0} e^{-jkz} & 0 \\ -\frac{E_{y0}^*}{\eta} e^{jkz} & \frac{E_{x0}^*}{\eta} e^{jkz} & 0 \end{vmatrix} = \left(\frac{E_{x0} E_{x0}^*}{\eta} + \frac{E_{y0}^* E_{y0}}{\eta} \right) \hat{\mathbf{z}}$$

$$= \frac{1}{\eta} \left(|E_{x0}|^2 + |E_{y0}|^2 \right) \hat{\mathbf{z}} = \frac{1}{\eta} |\tilde{\mathbf{E}}|^2 \hat{\mathbf{z}}. \tag{43}$$

Therefore,

$$\mathbf{S}_{\text{av}} = \frac{1}{2} \operatorname{Re} \left\{ \tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^* \right\} = \frac{1}{2\eta} |\tilde{\mathbf{E}}|^2 \hat{\mathbf{z}} \ \ (\text{W/m}^2). \tag{44}$$

Since the average power density only depends on η and the magnitude of $\tilde{\mathbf{E}}$, waves with different polarizations but the same magnitude carry the same power.

Electromagnetic Power Density (Continued)

Power of an EM Wave in a Lossy Medium

In this case, for a uniform plane wave traveling in the $+z$ direction,

$$\tilde{\mathbf{E}}(z) = (E_{x0}\hat{\mathbf{x}} + E_{y0}\hat{\mathbf{y}})e^{-\alpha z}e^{-j\beta z} \tag{45}$$

$$\tilde{\mathbf{H}}(z) = \frac{1}{\eta_c}(-E_{y0}\hat{\mathbf{x}} + E_{x0}\hat{\mathbf{y}})e^{-\alpha z}e^{-j\beta z} \tag{46}$$

So,

$$\begin{aligned}\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^* &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ E_{x0}e^{-\alpha z}e^{-j\beta z} & E_{y0}e^{-\alpha z}e^{-j\beta z} & 0 \\ -\frac{E_{y0}^*}{\eta_c^*}e^{-\alpha z}e^{j\beta z} & \frac{E_{x0}^*}{\eta_c^*}e^{-\alpha z}e^{j\beta z} & 0 \end{vmatrix} \\ &= \left(\frac{E_{x0}E_{x0}^*}{\eta_c^*}e^{-2\alpha z} + \frac{E_{y0}^*E_{y0}}{\eta_c^*}e^{-2\alpha z}\right)\hat{\mathbf{z}} = \frac{1}{\eta_c^*}|\tilde{\mathbf{E}}(0)|^2 e^{-2\alpha z}\hat{\mathbf{z}}. \end{aligned} \tag{47}$$

Electromagnetic Power Density (Continued)

Therefore,

$$\mathbf{S}_{\text{av}} = \frac{1}{2}\,\text{Re}\left\{\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*\right\} = \frac{1}{2}|\tilde{\mathbf{E}}(0)|^2 e^{-2\alpha z}\text{Re}\left\{\frac{1}{\eta_c^*}\right\}\hat{\mathbf{z}} \quad (\text{W/m}^2). \tag{48}$$

With

$$\eta_c = |\eta_c| e^{j\theta_\eta}, \tag{49}$$

$$\eta_c^* = |\eta_c| e^{-j\theta_\eta}, \quad \frac{1}{\eta_c^*} = \frac{1}{|\eta_c|} e^{j\theta_\eta} = \frac{1}{|\eta_c|}(\cos\theta_\eta + j\sin\theta_\eta),$$

$$\text{Re}\left\{\frac{1}{\eta_c^*}\right\} = \frac{1}{|\eta_c|}\cos\theta_\eta, \tag{50}$$

and

$$\mathbf{S}_{\text{av}} = \frac{|\tilde{\mathbf{E}}(0)|^2}{2|\eta_c|} e^{-2\alpha z}\cos\theta_\eta \hat{\mathbf{z}} \quad (\text{W/m}^2). \tag{51}$$

Electromagnetic Power Density (Continued)

Note that the only quantity in the average power density

$$\mathbf{S}_{\text{av}} = \frac{|\tilde{\mathbf{E}}(0)|^2}{2|\eta_c|} e^{-2\alpha z} \cos\theta_\eta \hat{\mathbf{z}} \quad (\text{W/m}^2) \tag{52}$$

that depends on the depth z is the exponential factor $e^{-2\alpha z}$ and that the average power density decreases at twice the rate 2α as the electric and magnetic fields

$$\tilde{\mathbf{E}}(z) = (E_{x0}\hat{\mathbf{x}} + E_{y0}\hat{\mathbf{y}})e^{-\alpha z}e^{-j\beta z} \quad (\text{V/m}) \tag{53}$$

$$\tilde{\mathbf{H}}(z) = \frac{1}{\eta_c}(-E_{y0}\hat{\mathbf{x}} + E_{x0}\hat{\mathbf{y}})e^{-\alpha z}e^{-j\beta z} \quad (\text{A/m}), \tag{54}$$

which both decrease at the rate α.

Electromagnetic Power Density (Continued)

Decibel Scale for Power Ratios

The ratio P_1/P_2 of two power levels P_1 and P_2 is often useful in applications. The most common situation is when a **measured power** P_{meas} is compared to a **reference power** P_0 through the ratio P_{meas}/P_0.

When the ratio

$$G = P_1/P_2 \tag{55}$$

is likely to have **values over several orders of magnitude**, such as when the power is related to the perceived value associated with a sensory system, and since **power is always non-negative**, it is helpful to express the ratio in terms of **decibels (dB)**:

$$G\ (\text{dB}) = 10 \log G = 10 \log \left(\frac{P_1}{P_2} \right). \tag{56}$$

Suppose $P_1 = V_1^2/R$ is the power dissipated in a resistor when voltage V_1 is applied across it and $P_2 = V_2^2/R$ is the power dissipated in the same resistor when voltage V_2 is applied across it.

Electromagnetic Power Density (Continued)

Then G when expressed in dB is

$$G\ (\text{dB}) = 10\log\left(\frac{P_1}{P_2}\right) = 10\log\left(\frac{V_1^2/R}{V_2^2/R}\right) = 10\log\left(\frac{V_1}{V_2}\right)^2$$
$$= 20\log\left(\frac{V_1}{V_2}\right) = 20\log g, \tag{57}$$

where $g = V_1/V_2$. So, in this case, the power ratio can be determined using the voltage ratio.

The rate of decrease of $S_{\text{av}}(z)$ as a function of depth z is called the **attenuation rate** and is defined as

$$A = 10\log\left(\frac{S_{\text{av}}(z)}{S_{\text{av}}(0)}\right) = 10\log\left(e^{-2\alpha z}\right) = -20\alpha z\log e = -8.69\alpha z, \tag{58}$$

where the unit of A is dB.

Electromagnetic Power Density (Continued)

If

$$\alpha \text{ (dB/m)} \equiv 8.69\alpha \text{ (Np/m)}, \tag{59}$$

then

$$A = -\alpha \text{ (dB/m)}\, z. \tag{60}$$

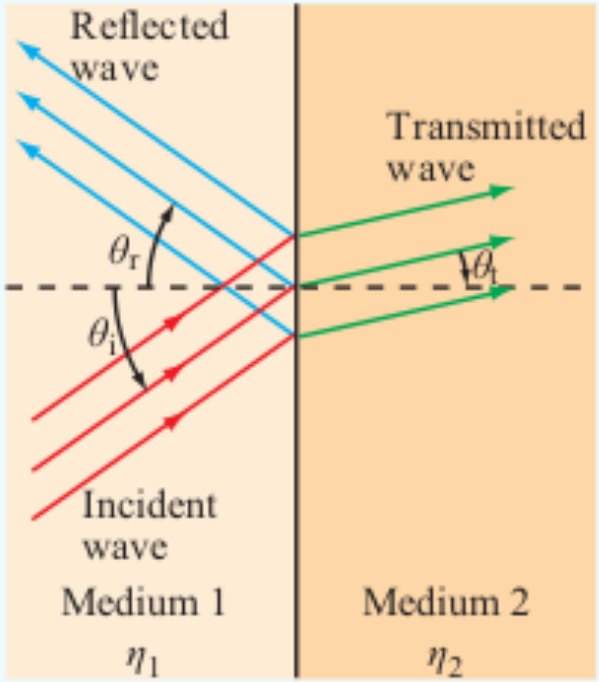
The attenuation rate can be written in terms of the magnitude of the electric field as

$$A = 10 \log \left(\frac{|E(z)|^2}{|E(0)|^2} \right) = 20 \log \left(\frac{|E(z)|}{|E(0)|} \right) \quad \text{(dB)}. \tag{61}$$

Chapter 8: Wave Reflection and Transmission

Suppose an EM wave is **traveling through Medium 1** and **reaches a boundary between Medium 1 and Medium 2**, where the two materials have **different electromagnetic properties**. When this occurs, the wave doesn't just continue traveling as it did in Medium 1. Instead,

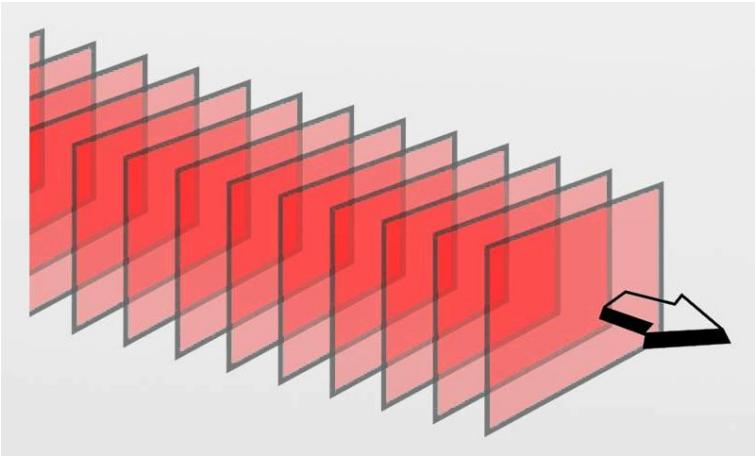
- a portion is **transmitted through the boundary** into Medium 2,
- another portion of the wave is **reflected back** into Medium 1,
- the **direction of propagation of the transmitted wave is different** from the original direction of propagation,
- the **speed of propagation is different** from the original speed of propagation.

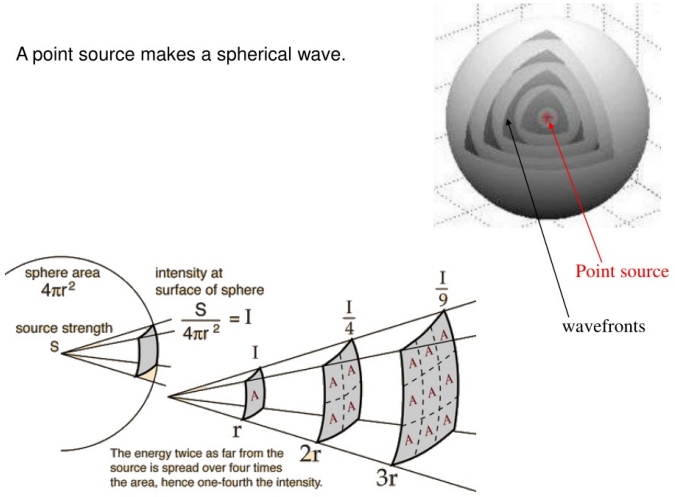


(b) Ray representation of oblique incidence

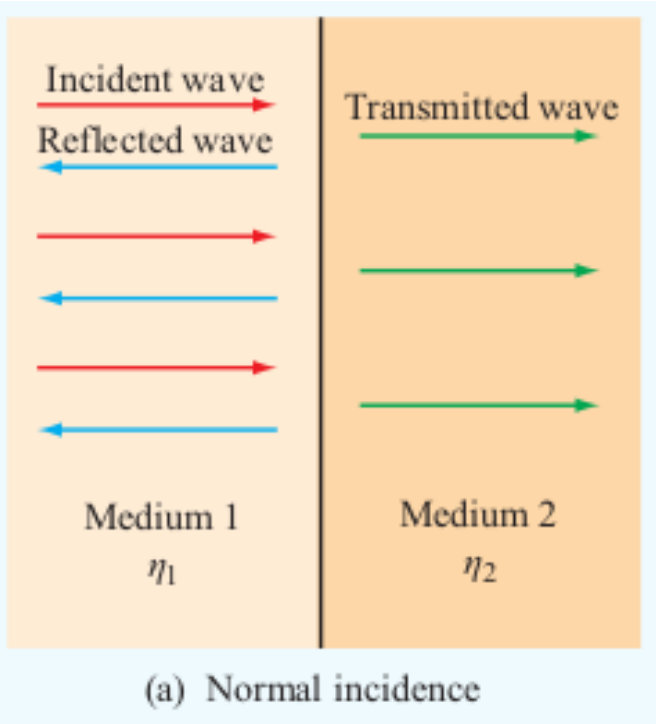
To understand this situation, it's useful to introduce some terminology.

- A **ray** is a line in the direction $\hat{\mathbf{k}}$ that an EM wave travels.
- A **wavefront** is a surface where the phase of a wave is constant.
- The wave traveling toward the boundary is the **incident wave**.
- The wave transmitted through the boundary is the **transmitted wave**.
- The wave reflected back into Medium 1 is the **reflected wave**.





If the **direction of propagation** in Medium 1 is the same as the **normal to the boundary**, then the transmitted and reflected waves travel in the same and opposite directions as the original wave, respectively.



(a) Normal incidence

This is analogous to what occurs at the boundary between two **transmission lines**. This allows the use of the methods developed for transmission lines in Chapter 2 in the context of a wave traveling normal to the boundary between two materials with different EM properties.

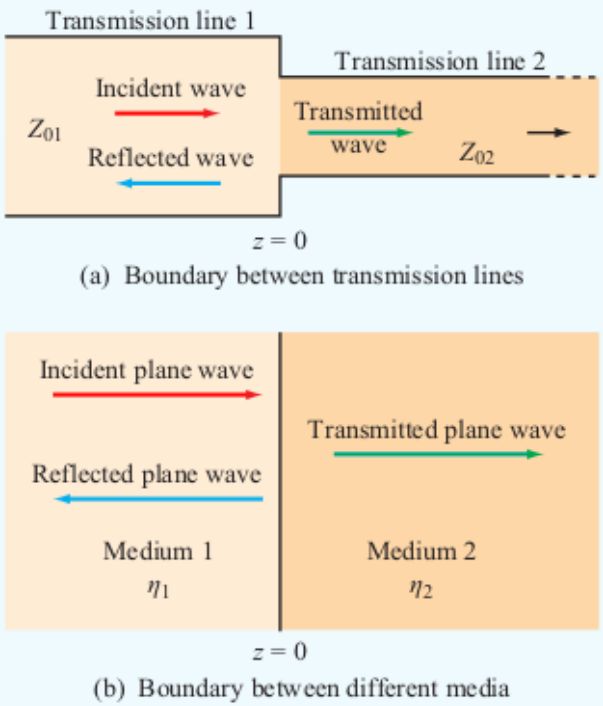


Figure 8-2 Discontinuity between two different transmission lines is analogous to that between two dissimilar media.

Reflection and Transmission: Normal Incidence

Boundary between Lossless Media

Medium 1

- Fills half-space $z \leq 0$
- Permittivity ϵ_1
- Permeability μ_1
- $k_1 = \omega\sqrt{\mu_1\epsilon_1}$, $\eta_1 = \sqrt{\mu_1/\epsilon_1}$

Medium 2

- Fills half-space $z \geq 0$
- Permittivity ϵ_2
- Permeability μ_2
- $k_2 = \omega\sqrt{\mu_2\epsilon_2}$, $\eta_2 = \sqrt{\mu_2/\epsilon_2}$

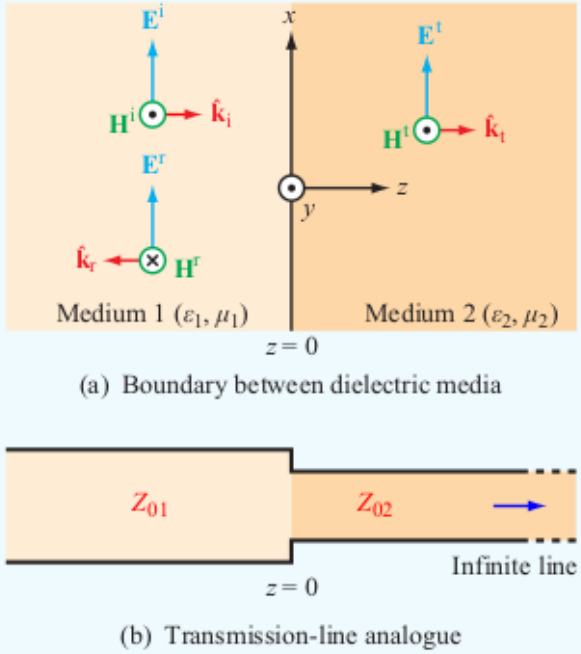


Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

Suppose that the incident wave is x-polarized with electric and magnetic fields $(\mathbf{E}^i, \mathbf{H}^i)$ and traveling in direction $\hat{\mathbf{k}}_i = \hat{\mathbf{z}}$.

Reflection and Transmission: Norm. Incidence (Continued)

Incident Wave $(\mathbf{E}^i, \mathbf{H}^i)$

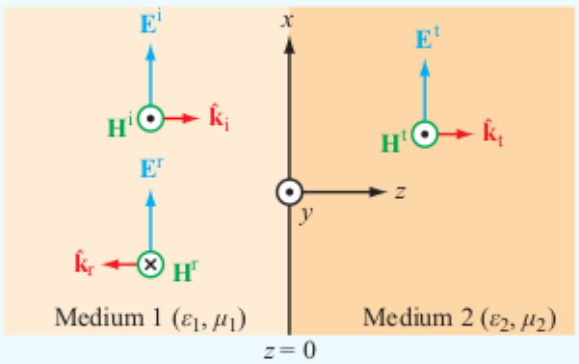
$$\tilde{\mathbf{E}}^i(z) = E_0^i e^{-jk_1 z}\hat{\mathbf{x}} \tag{62}$$

$$\tilde{\mathbf{H}}^i(z) = \hat{\mathbf{z}} \times \frac{\tilde{\mathbf{E}}^i(z)}{\eta_1} = \frac{E_0^i}{\eta_1} e^{-jk_1 z}\hat{\mathbf{y}} \tag{63}$$

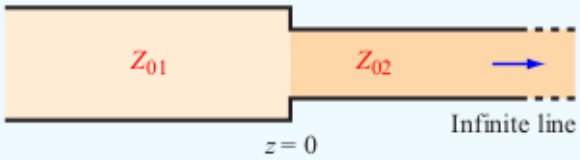
Reflected Wave $(\mathbf{E}^r, \mathbf{H}^r)$

$$\tilde{\mathbf{E}}^r(z) = E_0^r e^{jk_1 z}\hat{\mathbf{x}} \tag{64}$$

$$\tilde{\mathbf{H}}^r(z) = -(\hat{\mathbf{z}}) \times \frac{\tilde{\mathbf{E}}^r(z)}{\eta_1} = -\frac{E_0^r}{\eta_1} e^{jk_1 z}\hat{\mathbf{y}} \tag{65}$$



(a) Boundary between dielectric media



(b) Transmission-line analogue

Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

Reflection and Transmission: Norm. Incidence (Continued)

Transmitted Wave $(\mathbf{E}^t, \mathbf{H}^t)$

$$\tilde{\mathbf{E}}^t(z) = E_0^t e^{-jk_2 z}\hat{\mathbf{x}} \tag{66}$$

$$\tilde{\mathbf{H}}^t(z) = \hat{\mathbf{z}} \times \frac{\tilde{\mathbf{E}}^t(z)}{\eta_2} = \frac{E_0^t}{\eta_2} e^{-jk_2 z}\hat{\mathbf{y}} \tag{67}$$

E_0^i, E_0^r, and E_0^t are the **amplitudes** of $\mathbf{E}^i$, $\mathbf{E}^r$, and $\mathbf{E}^t$ at $z = 0$, respectively. The relationships between these amplitudes is determine by **boundary conditions**.

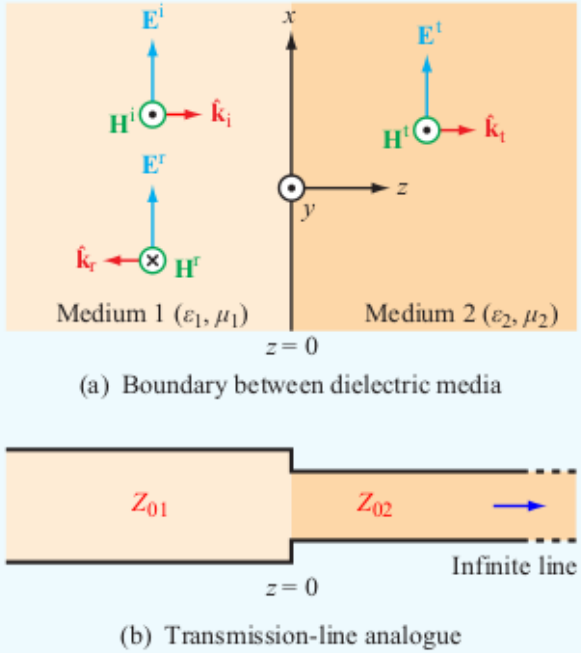


Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

Reflection and Transmission: Norm. Incidence (Continued)

Medium 1

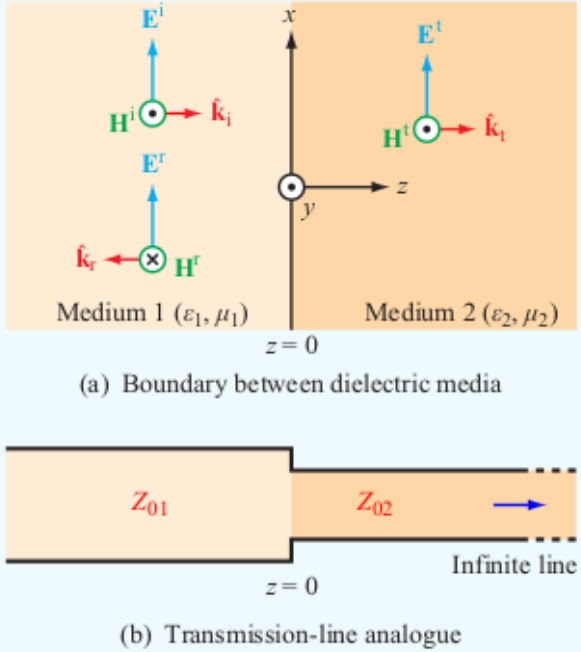
$$\tilde{\mathbf{E}}_1(z) = \tilde{\mathbf{E}}^i(z) + \tilde{\mathbf{E}}^r(z)$$
$$= \left(E_0^i e^{-jk_1 z} + E_0^r e^{jk_1 z}\right)\hat{\mathbf{x}} \quad (68)$$

$$\tilde{\mathbf{H}}_1(z) = \tilde{\mathbf{H}}^i(z) + \tilde{\mathbf{H}}^r(z)$$
$$= \frac{1}{\eta_1}\left(E_0^i e^{-jk_1 z} - E_0^r e^{jk_1 z}\right)\hat{\mathbf{y}} \quad (69)$$

Medium 2

$$\tilde{\mathbf{E}}_2(z) = \tilde{\mathbf{E}}^t(z) = E_0^t e^{-jk_2 z}\hat{\mathbf{x}} \quad (70)$$

$$\tilde{\mathbf{H}}_2(z) = \tilde{\mathbf{H}}^t(z) = \frac{E_0^t}{\eta_2} e^{-jk_2 z}\hat{\mathbf{y}} \quad (71)$$



(a) Boundary between dielectric media

(b) Transmission-line analogue

Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

Reflection and Transmission: Norm. Incidence (Continued)

The tangential components of **E** and **H** at $z = 0$ are continuous, i.e.

$$\tilde{\mathbf{E}}_1(0) = \tilde{\mathbf{E}}_2(0) \tag{72}$$

$$\tilde{\mathbf{H}}_1(0) = \tilde{\mathbf{H}}_2(0), \tag{73}$$

which means

$$E_0^i + E_0^r = E_0^t \tag{74}$$

$$\frac{1}{\eta_1}\left(E_0^i - E_0^r\right) = \frac{E_0^t}{\eta_2}. \tag{75}$$

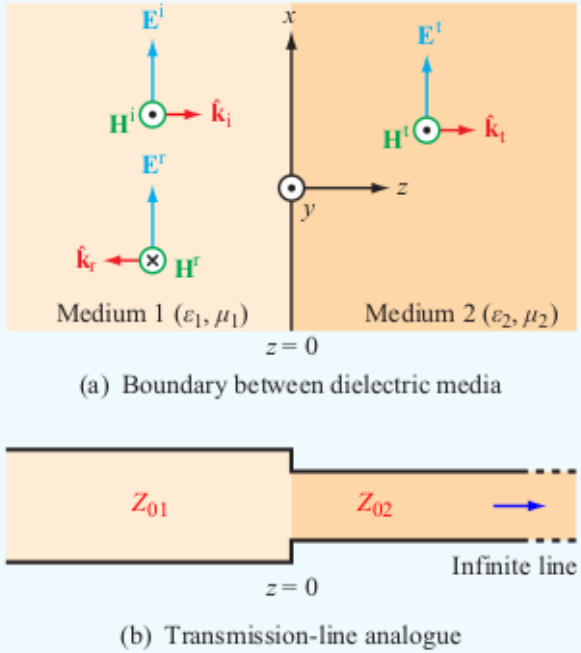


Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

Reflection and Transmission: Norm. Incidence (Continued)

The goal is to solve the equations

$$E_0^i + E_0^r = E_0^t, \quad \frac{1}{\eta_1}\left(E_0^i - E_0^r\right) = \frac{E_0^t}{\eta_2}. \tag{76}$$

for E_0^r and E_0^t in terms of E_0^i.

The 1st equation implies

$$E_0^r = E_0^t - E_0^i. \tag{77}$$

Multiplying the 2nd by η_1 and substituting in the expression for E_0^r,

$$\frac{\eta_1}{\eta_2} E_0^t = E_0^i - \left(E_0^t - E_0^i\right) = 2E_0^i - E_0^t \tag{78}$$

So,

$$\left(\frac{\eta_2 + \eta_1}{\eta_2}\right) E_0^t = \left(1 + \frac{\eta_1}{\eta_2}\right) E_0^t = 2E_0^i \implies E_0^t = \left(\frac{2\eta_2}{\eta_2 + \eta_1}\right) E_0^i. \tag{79}$$

Reflection and Transmission: Norm. Incidence (Continued)

Substituting this into the expression obtained from the 1st equation,

$$E_0^r = E_0^t - E_0^i = \left(\frac{2\eta_2}{\eta_2+\eta_1}\right) E_0^i - E_0^i = \left(\frac{2\eta_2}{\eta_2+\eta_1} - 1\right) E_0^i$$
$$= \left(\frac{2\eta_2 - (\eta_2+\eta_1)}{\eta_2+\eta_1}\right) E_0^i = \left(\frac{\eta_2-\eta_1}{\eta_2+\eta_1}\right) E_0^i. \tag{80}$$

Therefore, in summary,

$$E_0^r = \left(\frac{\eta_2-\eta_1}{\eta_2+\eta_1}\right) E_0^i = \Gamma E_0^i, \quad E_0^t = \left(\frac{2\eta_2}{\eta_2+\eta_1}\right) E_0^i = \tau E_0^i, \tag{81}$$

where the **reflection coefficient** is defined to be

$$\Gamma = \frac{\eta_2-\eta_1}{\eta_2+\eta_1} \tag{82}$$

and the **transmission coefficient** is defined to be

$$\tau = \frac{2\eta_2}{\eta_2+\eta_1}. \tag{83}$$

Reflection and Transmission: Norm. Incidence (Continued)

In this case, the intrinsic impedances are **real-valued** and so are Γ and τ. Furthermore,

$$1 + \Gamma = 1 + \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{\eta_2 + \eta_1 + \eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{2\eta_2}{\eta_2 + \eta_1} = \tau. \tag{84}$$

When the media are **nonmagnetic**, $\mu_1 = \mu_2 = \mu_0$,

$$\eta_1 = \frac{\eta_0}{\sqrt{\epsilon_{r1}}}, \quad \eta_2 = \frac{\eta_0}{\sqrt{\epsilon_{r2}}}, \tag{85}$$

and

$$\Gamma = \frac{\sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}}. \tag{86}$$

Homework

Read Chapter 7.

Homework 2 is posted and will be due on Tuesday, May 5.

Exam 1 will be Thursday, May 7.